

**Q Answer the following questions with justifications.**

- (1) Given the characteristic equation of a matrix  $A$ , can we compute the characteristic equation of  $cA$  where  $c$  is a non-zero scalar without knowing the entries of  $A$ ? If so, show how to do it using detailed calculations. Otherwise explain why it is not possible. Clearly state all your assumptions

(2 Marks)

- (2) Consider an inner product space with an inner product  $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{A} \mathbf{y}$  defined with the help of matrix  $\mathbf{A}$  defined below.

$$\mathbf{A} = \begin{bmatrix} 5.5 & -1.5 \\ -1.5 & 5.5 \end{bmatrix}$$

Consider two vectors  $\mathbf{a} = [1 \ 5]^T$  and  $\mathbf{b} = [2 \ 7]^T$  in the inner product space.

- (a) Find the distance  $d(\mathbf{a}, \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|$  between vectors  $\mathbf{a}$  and  $\mathbf{b}$  in the above inner product space where  $\|\cdot\|$  is the norm induced by the inner product.

(2 marks)

- (b) Find the angle between vectors  $\mathbf{a}$  and  $\mathbf{b}$  in the above inner product space.

(2 marks)

**A Answers**

- (1) The characteristic equation of the matrix  $cA$  is  $\det(cA - \lambda I) = 0$ . We can rewrite this as  $\det(c(A - \frac{\lambda}{c}I)) = 0$  which can then be written as  $c^n \det(A - \frac{\lambda}{c}I) = 0$  or  $\det(A - \frac{\lambda}{c}I) = 0$ . Let  $\mu = \frac{\lambda}{c}$ , and assume that the characteristic equation  $\det(A - \mu I) = 0$  can be written in terms of the polynomial  $\mu^n + a_{n-1}\mu^{n-1} + \dots + a_1\mu + a_0 = 0$ . Substituting  $\lambda/c = \mu$  in this polynomial we get  $(\frac{\lambda}{c})^n + a_{n-1}(\frac{\lambda}{c})^{n-1} + \dots + a_1(\frac{\lambda}{c}) + a_0 = 0$ . Multiplying through by  $c^n$  we finally get  $\lambda^n + ca_{n-1}\lambda^{n-1} + \dots + a_1c^{n-1}\lambda + a_0c^n = 0$ . Thus we see that the characteristic equation of  $cA$  can be obtained by taking the coefficient  $a_k$  of the  $k$ th term of the characteristic equation of  $A$  and multiplying it by  $c^{n-k}$ . Thus there is no need to look at the entries of the matrix  $A$ .

Marking Scheme: 1 Mark  $\rightarrow$  expanding the determinant  $\det(A - \frac{\lambda}{c}I)$ , and obtaining the equation  $\lambda^n + ca_{n-1}\lambda^{n-1} + \dots + a_1c^{n-1}\lambda + a_0c^n = 0$ . 1 Mark  $\rightarrow$  remaining argument.

- (2) (a)

$$(d(\mathbf{a}, \mathbf{b}))^2 = [1 \ -2 \ 5 \ -7] \begin{bmatrix} 5.5 & -1.5 \\ -1.5 & 5.5 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = 21.5$$

$$d(\mathbf{a}, \mathbf{b}) = 4.63$$

(b)

$$\cos(\theta) = \frac{\mathbf{a}^T \mathbf{A} \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{\mathbf{a}^T \mathbf{A} \mathbf{b}}{\sqrt{\mathbf{a}^T \mathbf{A} \mathbf{a}} \sqrt{\mathbf{b}^T \mathbf{A} \mathbf{b}}}$$
$$\cos(\theta) = \frac{178}{\sqrt{128} \sqrt{249.5}} = 0.005573$$

$$\theta = 89.6 \text{ deg}$$

Marking Scheme: 1 mark each for proper construction of equation.  
1 mark for correct numerical answer

**Q Answer the following for the given matrix:**

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

(A) Using elementary row operations, write the matrix in its row echelon form

(2 marks)

(B) Let  $\mathbf{V}$  be a vector subspace spanned by the columns of matrix  $\mathbf{A}$ . Find the basis and dimension of  $\mathbf{V}$ .

(2 marks)

(C) Let  $\mathbf{V}$  be a vector subspace spanned by vectors  $\mathbf{x}$ , such that  $\mathbf{Ax} = \mathbf{0}$ . Find the basis and dimension of  $\mathbf{V}$ .

(2 marks)

(D) Give the set of linearly independent rows of  $\mathbf{A}$ . What is the number of vectors in this set?

(2 marks)

**A Answers**

(A)

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 3 * R_1, R_3 \leftarrow R_3 - 2 * R_1, R_2 \leftarrow R_2 / 5, R_3 \leftarrow R_3 - R_2, R_1 \leftarrow R_1 - 2 * R_2}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

NOTE: REF is not unique.

1 Mark for correct elementary row operations, 1 Mark for getting correct pivot columns

(B) column number 1 and column number 3 have pivot.

Basis of column space is:

$$\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

(1 mark)

Rank of col space = dimension of  $\mathbf{V} = 2$

(1 mark)

(C) From REF, the null space is:

$$\begin{bmatrix} x \\ r \\ y \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} t$$

basis =

$$\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

(1 mark)

Rank of null space = dimension of  $\mathbf{V} = 3$

(1 mark)

(D) From REF, row 1 and row 2 form basis of row space

(1 mark)

Rank of row space = 2

(1 mark)

**Q Answer the following for the given matrix:**

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(A) Obtain the left-singular vectors of  $\mathbf{A}$ .

(3 marks)

(B) Obtain the right-singular vectors of  $\mathbf{A}$ .

(3 marks)

(C) Obtain the singular value matrix  $\mathbf{\Sigma}$ . What is the spectral norm of  $\mathbf{A}$ ?

(2 marks)

**A Answers**

(A)

$$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{A}^T = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \mathbf{A}\mathbf{A}^T = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

eigenvalues and eigenvectors of  $\mathbf{A}\mathbf{A}^T$  :

$$\lambda_1 = \sigma_1^2 = 6, u_1 = [5 \quad 2 \quad 1]^T, \|u_1\| = \sqrt{30}$$

$$\lambda_2 = \sigma_2^2 = 1, u_2 = [0 \quad \frac{-1}{2} \quad 1]^T, \|u_2\| = \frac{\sqrt{5}}{2}$$

$$\lambda_3 = \sigma_3^2 = 0, u_3 = [-1 \quad 2 \quad 1]^T, \|u_3\| = \sqrt{6}$$

Singular value matrix:

$$\mathbf{\Sigma} = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & \sqrt{1} \\ 0 & 0 \end{bmatrix}$$

Left singular vectors

$$\mathbf{U} = \begin{bmatrix} \frac{u_1}{\|u_1\|} & \frac{u_2}{\|u_2\|} & \frac{u_3}{\|u_3\|} \end{bmatrix}$$

Marking Scheme: 1 mark for getting eigenvalues, 1 mark for eigenvectors, 1 mark for construction of  $\mathbf{U}$

(B) Right Singular vectors:

$$v_1 = \frac{1}{\sigma_1} \mathbf{A}^T u_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} \frac{1}{\sqrt{30}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \frac{6}{\sqrt{6}\sqrt{30}}$$

$$v_2 = \frac{1}{\sigma_2} \mathbf{A}^T u_2 = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{2} \\ 1 \end{bmatrix} \frac{2}{\sqrt{5}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \frac{1}{\sqrt{5}}$$

Marking Scheme: 1 mark for getting eigenvalues, 1 mark for eigenvectors, 1 mark for construction of  $\mathbf{U}$

(C) Singular value matrix:

$$\mathbf{\Sigma} = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & \sqrt{1} \\ 0 & 0 \end{bmatrix}$$

Spectral norm =  $\sqrt{6}$  Marking Scheme: 1 mark for matrix, 1 mark for spectral norm

**Q** Answer the following

(1) Compute the following for the function  $f(x_1, x_2) = e^{x_1} + x_1 x_2 - \log(1 + x_2)$ .

(A) The expression for gradient, its dimension and its value at  $(1, 2)$ .  
(2 marks)

(B) The expression for the Hessian matrix, its dimension and its value at  $(1, 2)$ .  
(2 marks)

(C) The derivative  $\frac{df}{dt}$  using chain rule of differentiation when  $x_1 = t^2 + 2at$ ,  $x_2 = \sin(t)$ .  
(2 marks)

(2) Define  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  as  $f(\mathbf{x}) = (\mathbf{A}\mathbf{x} - \mathbf{b})^T(\mathbf{A}\mathbf{x} - \mathbf{b})$  where  $\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix}$ ,  $\mathbf{b} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$ . Find Taylor's polynomial of degree 1 of  $f$  at  $[1, 1]$ .  
(2 Marks)

**A Answers**

(1) (A)

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} e^{x_1} + x_2 & x_1 - \frac{1}{1+x_2} \end{bmatrix}$$

$$\nabla_{\mathbf{x}} f(1, 2) = \begin{bmatrix} e^1 + 2 & \frac{2}{3} \end{bmatrix}$$

Marking Scheme: 1 mark for expression, 1 mark for value at  $(1, 2)$

(B)

$$\nabla_{\mathbf{x}}^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} e^{x_1} & 1 \\ 1 & \frac{1}{(1+x_2)^2} \end{bmatrix}$$

$$\nabla_{\mathbf{x}}^2 f(1, 2) = \begin{bmatrix} e^1 & 1 \\ 1 & \frac{1}{9} \end{bmatrix}$$

Marking Scheme: 1 mark for expression, 1 mark for value at  $(1, 2)$

(C)

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}$$

$$= (e^{x_1} + x_2)(2t + 2a) + (x_1 - \frac{1}{1+x_2})(\cos t)$$

$$= (e^{(t^2+2at)} + \sin t)(2t + 2a) + (t^2 + 2at - \frac{1}{1+\sin(t)}) \cos t$$

Marking Scheme: 1 mark for chain rule expression, 1 mark for final solution

- (2) Using the identity, we get  $\nabla_{\mathbf{x}} f(\mathbf{x}) = 2(\mathbf{A}\mathbf{x} - \mathbf{b})^T \mathbf{A}$ .  
Therefore, we have

$$\begin{aligned} f([1, 1]^T) &= 2 \\ \nabla_{\mathbf{x}} f([1, 1]^T) &= [10, 10] \end{aligned}$$

Therefore, if  $\mathbf{x} = [x_1, x_2]^T$ , Taylor's Polynomial is defined as

$$\begin{aligned} T_1 f(\mathbf{x}) &= f([1, 1]^T) + \nabla_{\mathbf{x}} f([1, 1]^T) [(x_1 - 1), (x_2 - 1)]^T, \\ &= 10x_1 + 10x_2 - 18 \end{aligned}$$

Marking scheme: 0.5 Marks for computing  $f([1, 1]^T)$ , 2 Marks for computing  $\nabla_{\mathbf{x}} f$  and 0.5 Marks for  $\nabla_{\mathbf{x}} f([1, 1]^T)$ . 1 Mark for the computing Taylor's polynomial.